

06026240 INTELLIGENT SYSTEM DEVELOPMENT

Chapter V Generative adversarial network (GAN)

with Dr. Bhattarabhorn Wattanacheep

- Generative modeling — การเรียนรู้การกระจายของข้อมูล
- Autoencoders (AE)
- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs)
- GANs — ความก้าวหน้าและการประยุกต์ใช้
- Lab — สร้างภาพใบหน้าแมวด้วย Generative Model

Supervised vs unsupervised learning

Supervised Learning

Data: (x, y)

x is data, y is label

Goal: Learn function to map
 $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, etc.

Unsupervised Learning

Data: x

x is data, no labels!

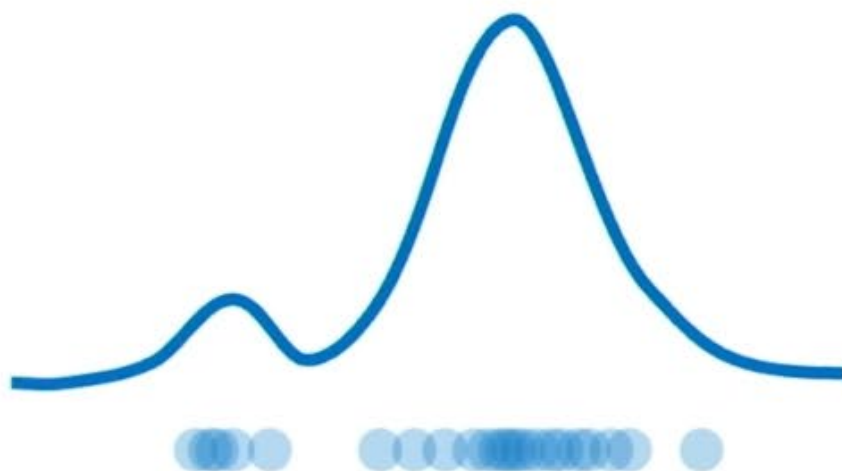
Goal: Learn the *hidden* or *underlying structure* of the data

Examples: Clustering, feature or dimensionality reduction, etc.

Generative modeling

Goal: Take as input training samples from some distribution and learn a model that represents that distribution

Density Estimation

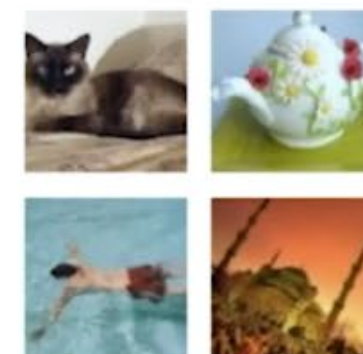


Sample Generation



Input samples

Training data $\sim P_{data}(x)$

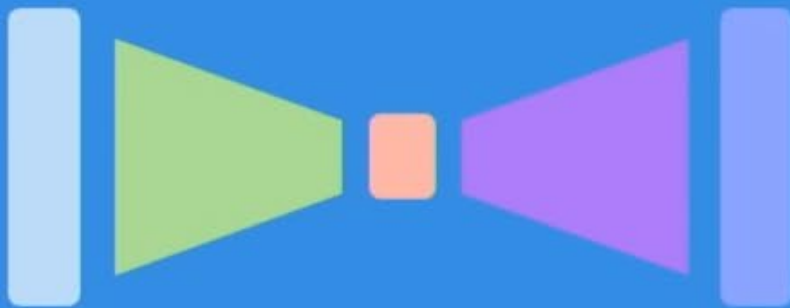


Generated samples

Generated $\sim P_{model}(x)$

Latent variable models

Autoencoders and Variational
Autoencoders (VAEs)



Generative Adversarial
Networks (GANs)



What is a latent variable?

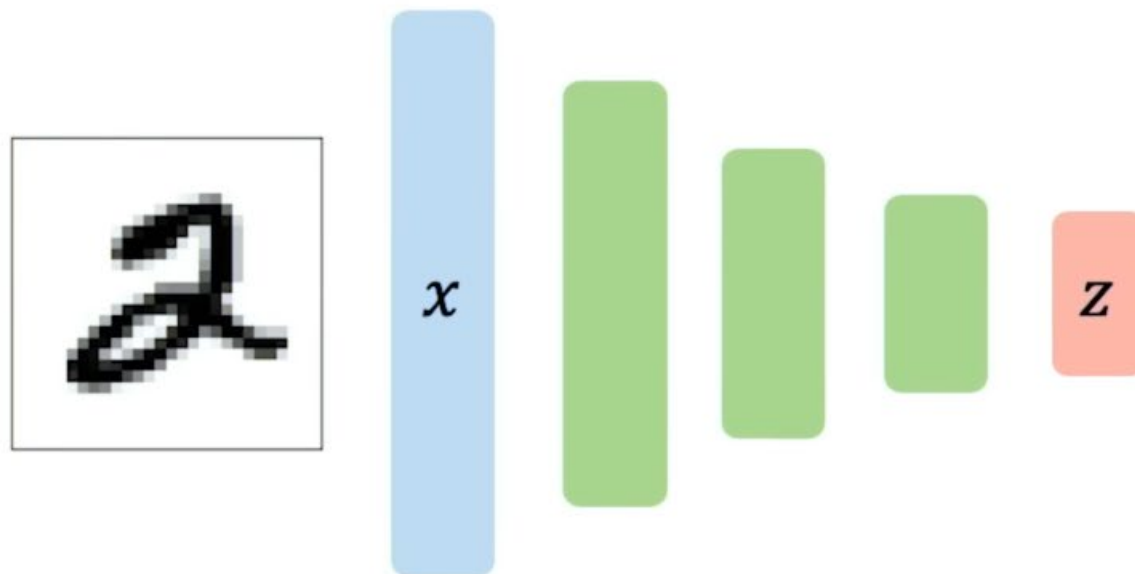


Myth of the Cave

Autoencoders

Autoencoders: background

Unsupervised approach for learning a **lower-dimensional** feature representation from unlabeled training data



Why do we care about a low-dimensional z ?

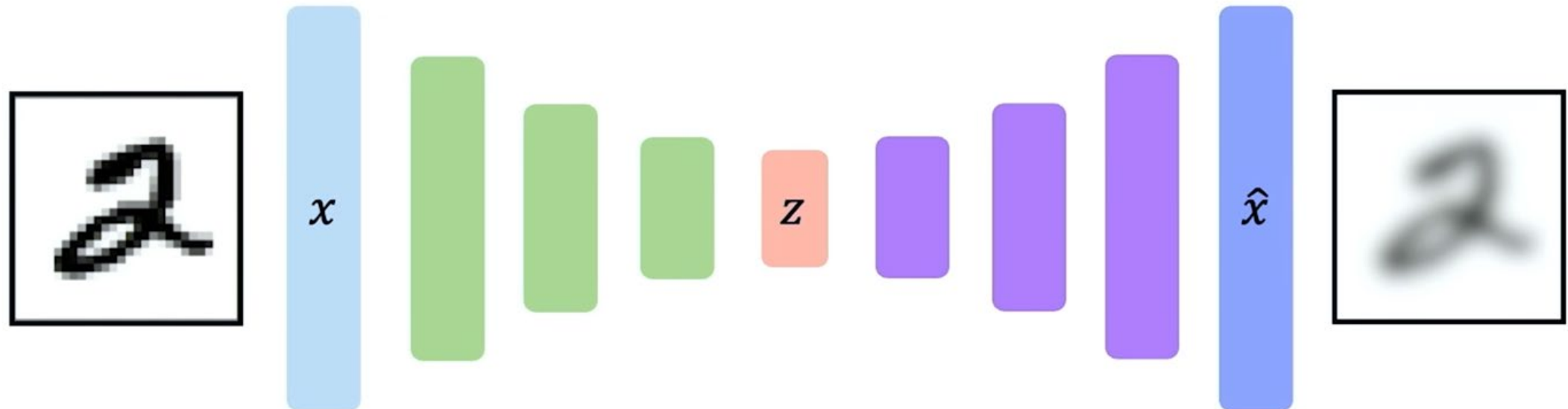


“Encoder” learns mapping from the data, x , to a low-dimensional latent space, z

Autoencoders: background

How can we learn this latent space?

Train the model to use these features to **reconstruct the original data**

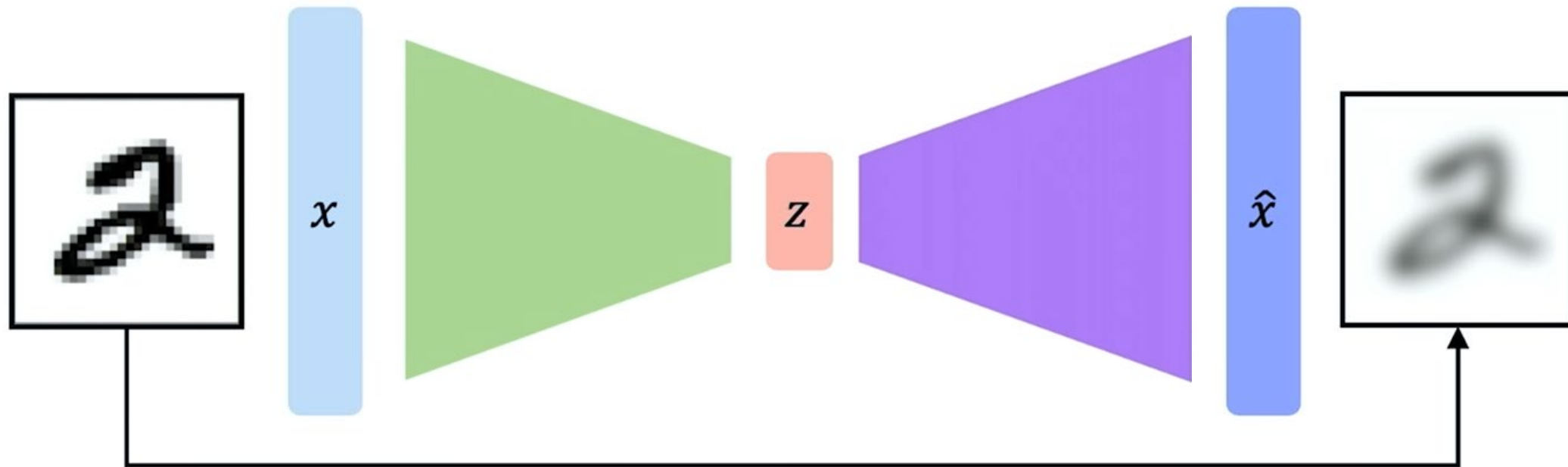


“Decoder” learns mapping back from latent space, z ,
to a reconstructed observation, $\hat{x}$

Autoencoders: background

How can we learn this latent space?

Train the model to use these features to **reconstruct the original data**



$$\mathcal{L}(x, \hat{x}) = \|x - \hat{x}\|^2$$

Loss function doesn't
use any labels!!

Dimensionality of latent space $\rightarrow$ reconstruction quality

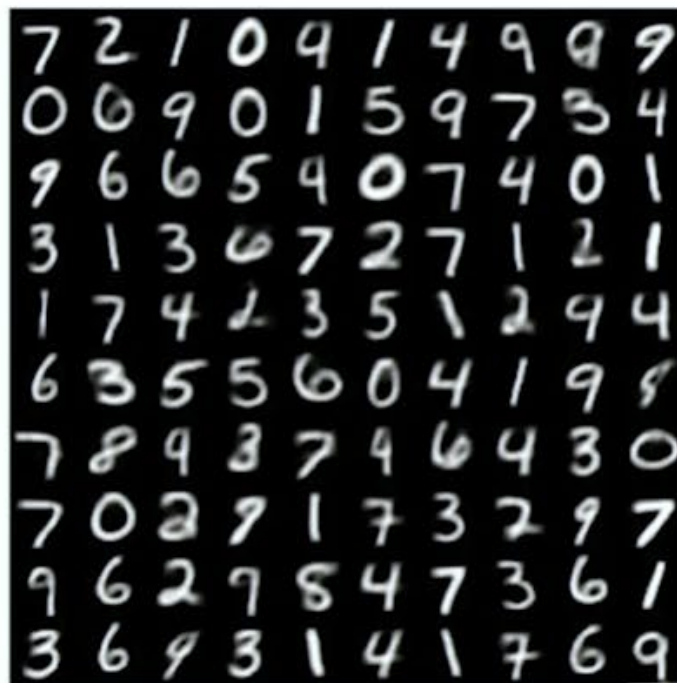
Autoencoding is a form of compression!

Smaller latent space will force a larger training bottleneck

2D latent space



5D latent space



Ground Truth



Autoencoders for representation learning

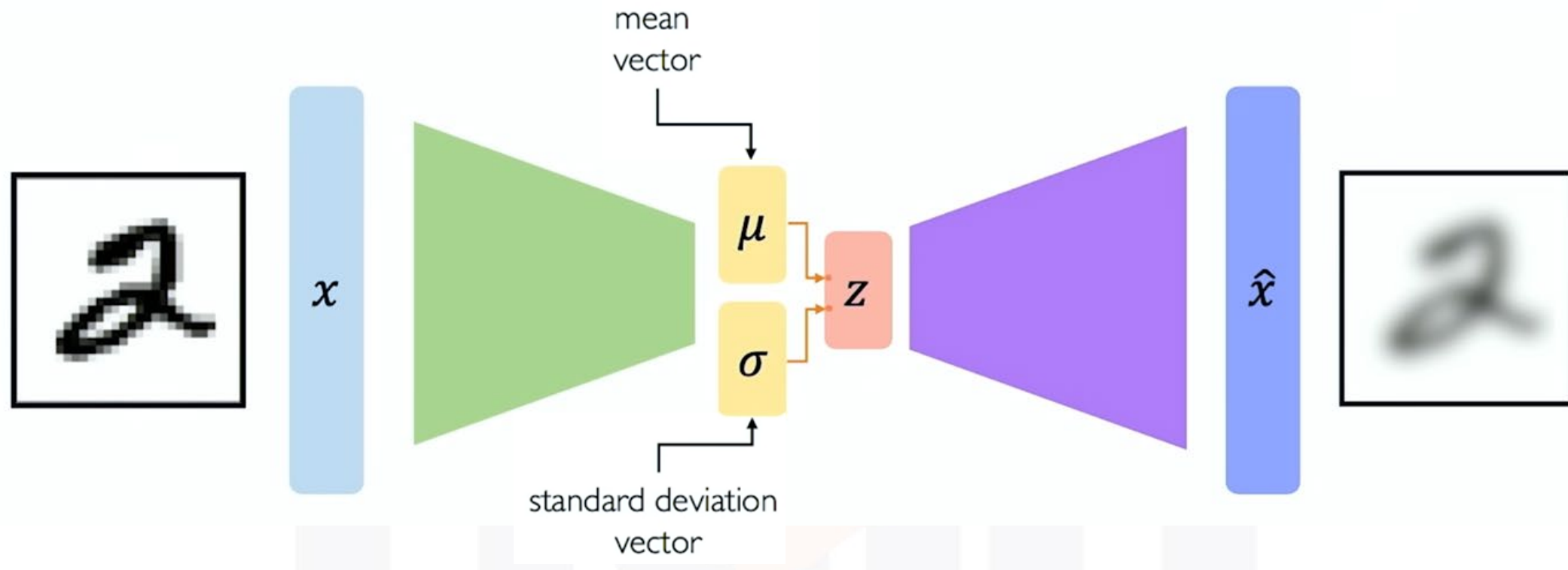
Bottleneck hidden layer forces network to learn a compressed latent representation

Reconstruction loss forces the latent representation to capture (or encode) as much “information” about the data as possible

Autoencoding = **Auto** automatically **encoding** data; “Auto” = **self**-encoding

Variational Autoencoders (VAEs)

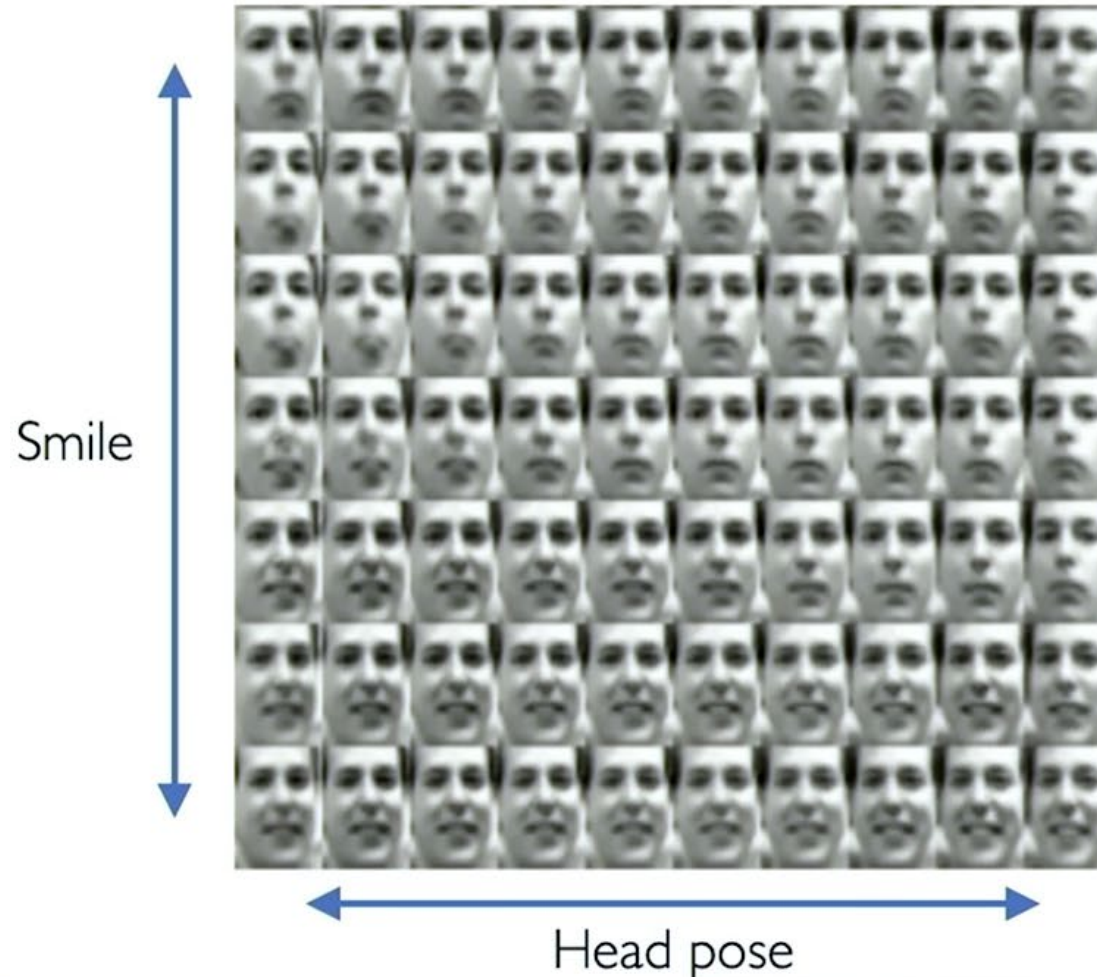
VAEs: key difference with traditional autoencoder



Variational autoencoders are a probabilistic twist on autoencoders!

Sample from the mean and standard deviation to compute latent sample

VAEs: Latent perturbation



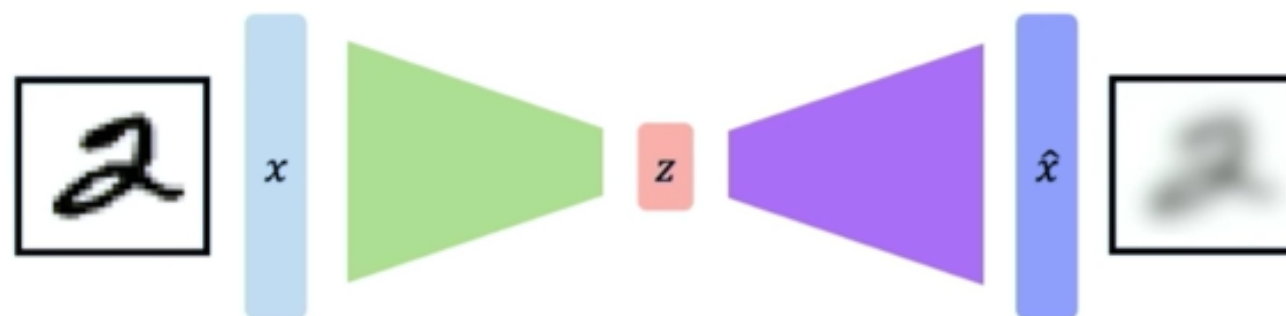
Ideally, we want latent variables that are uncorrelated with each other

Enforce diagonal prior on the latent variables to encourage independence

Disentanglement

VAE summary

1. Compress representation of world to something we can use to learn
2. Reconstruction allows for unsupervised learning (no labels!)
3. Reparameterization trick to train end-to-end
4. Interpret hidden latent variables using perturbation
5. Generating new examples

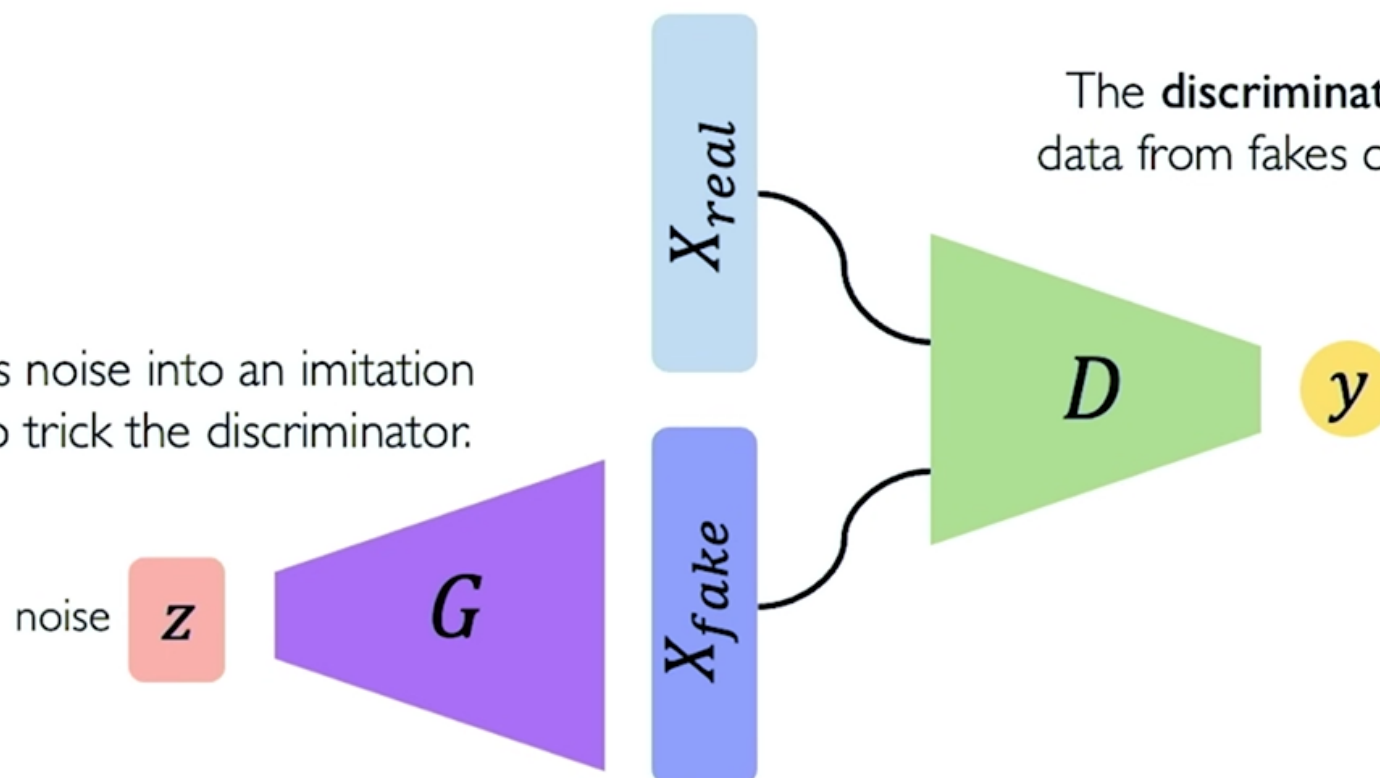


Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs)

Generative Adversarial Networks (GANs) are a way to make a generative model by having two neural networks compete with each other.

The **generator** turns noise into an imitation of the data to try to trick the discriminator.



The **discriminator** tries to identify real data from fakes created by the generator.

Intuition behind GANs

Discriminator looks at both real data and fake data created by the generator.

Discriminator

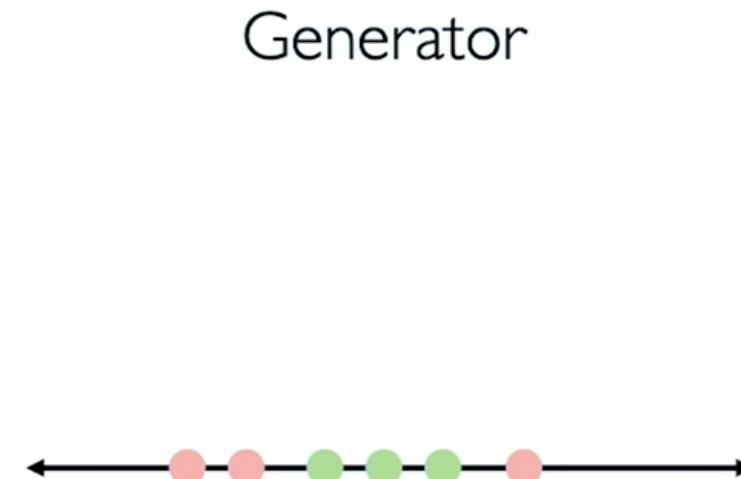
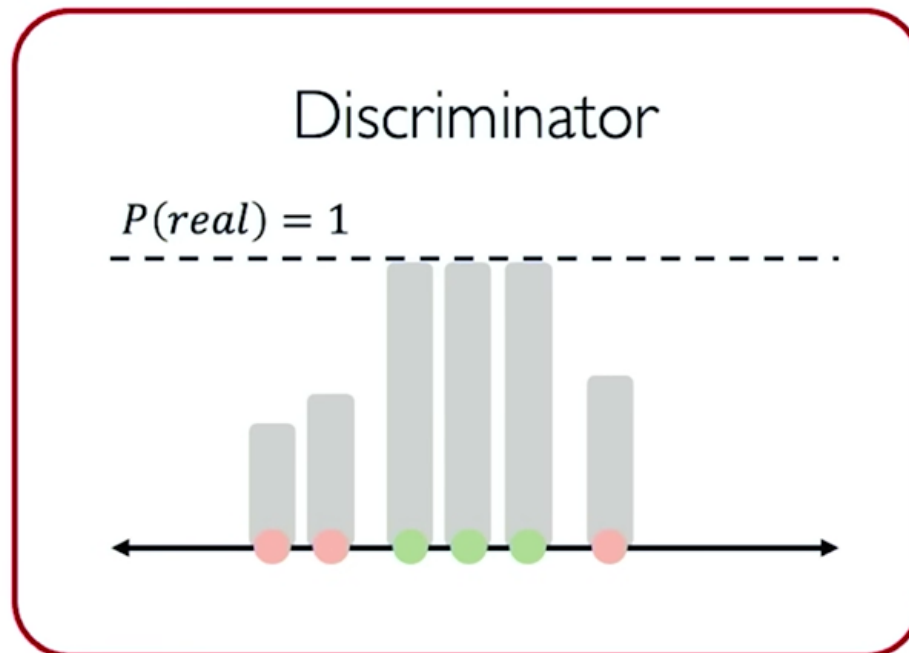
Generator



 Fake data

Intuition behind GANs

Discriminator tries to predict what's real and what's fake.



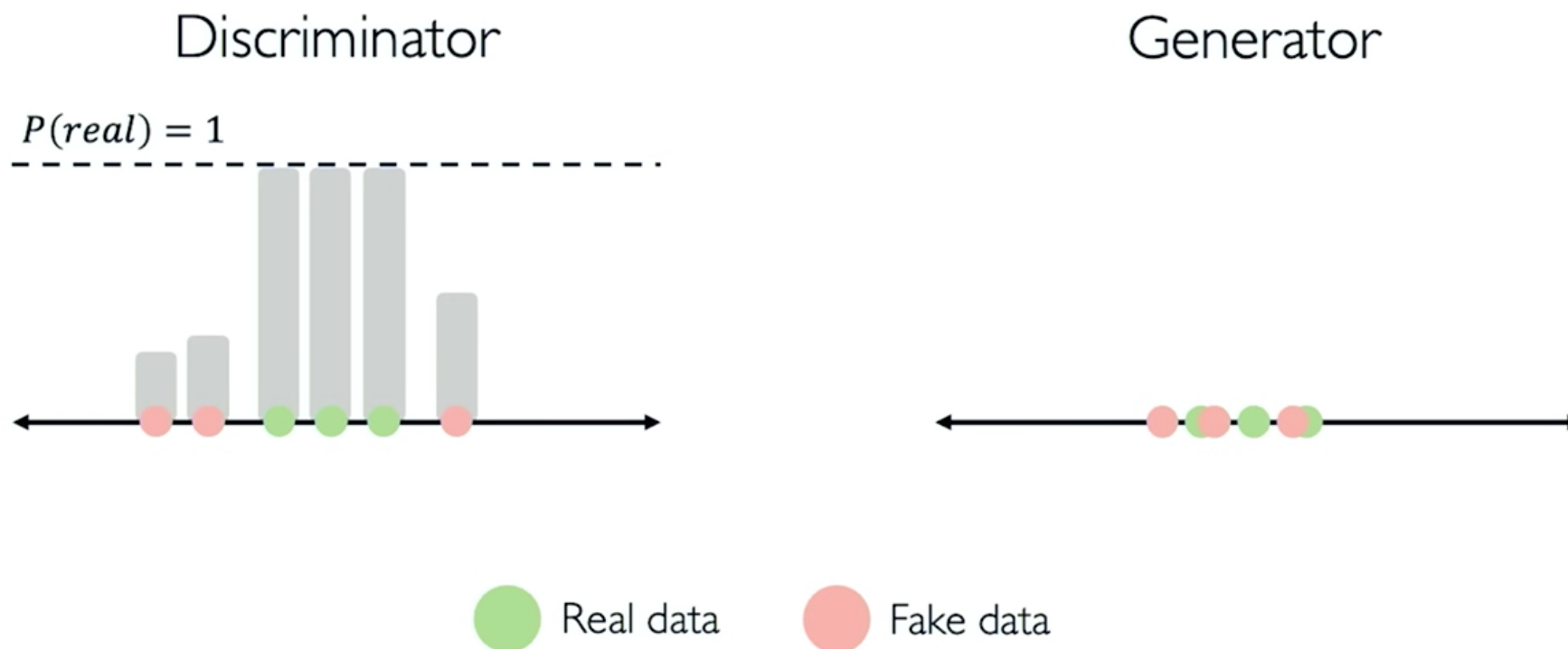
● Real data

● Fake data

Intuition behind GANs

Discriminator tries to identify real data from fakes created by the generator.

Generator tries to create imitations of data to trick the discriminator.



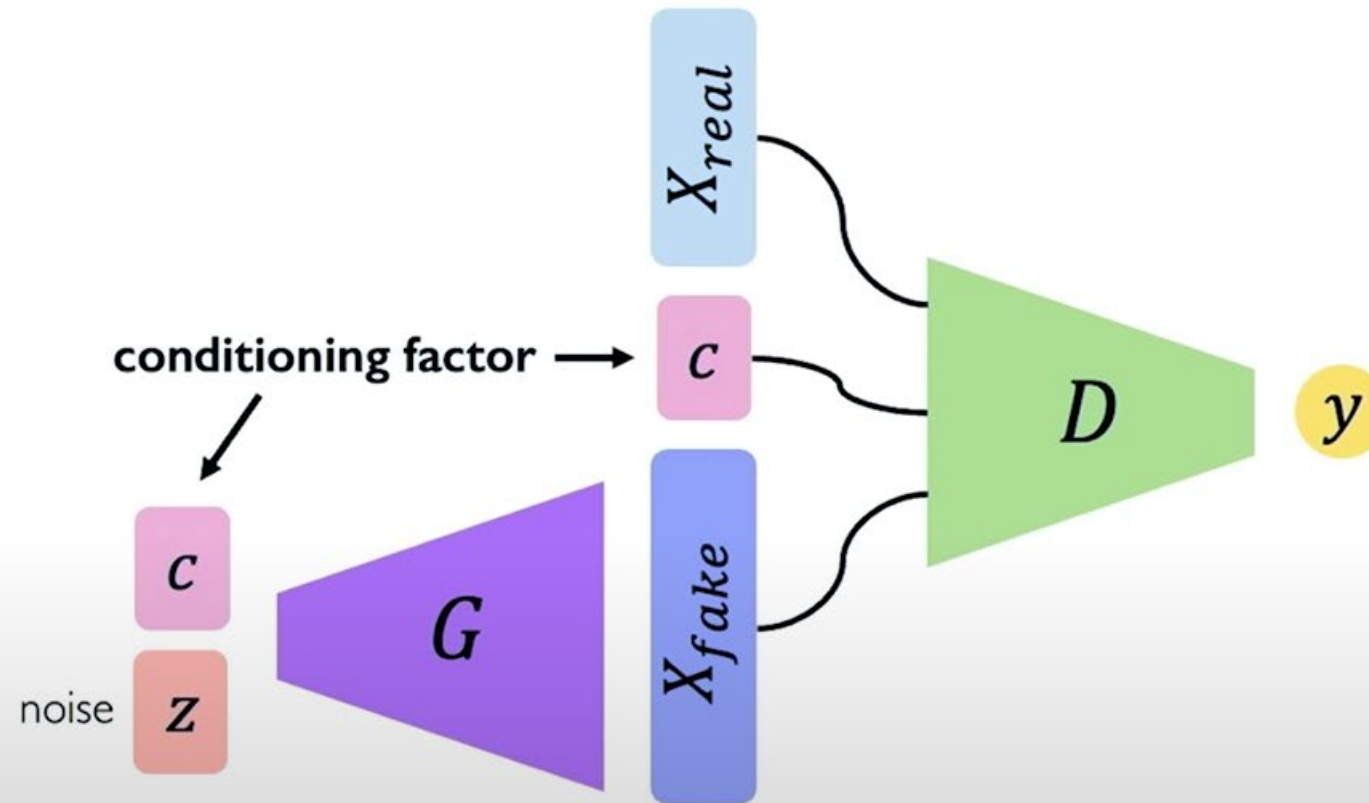
GANs: Advances and Applications

Progressive growing of GANs: results

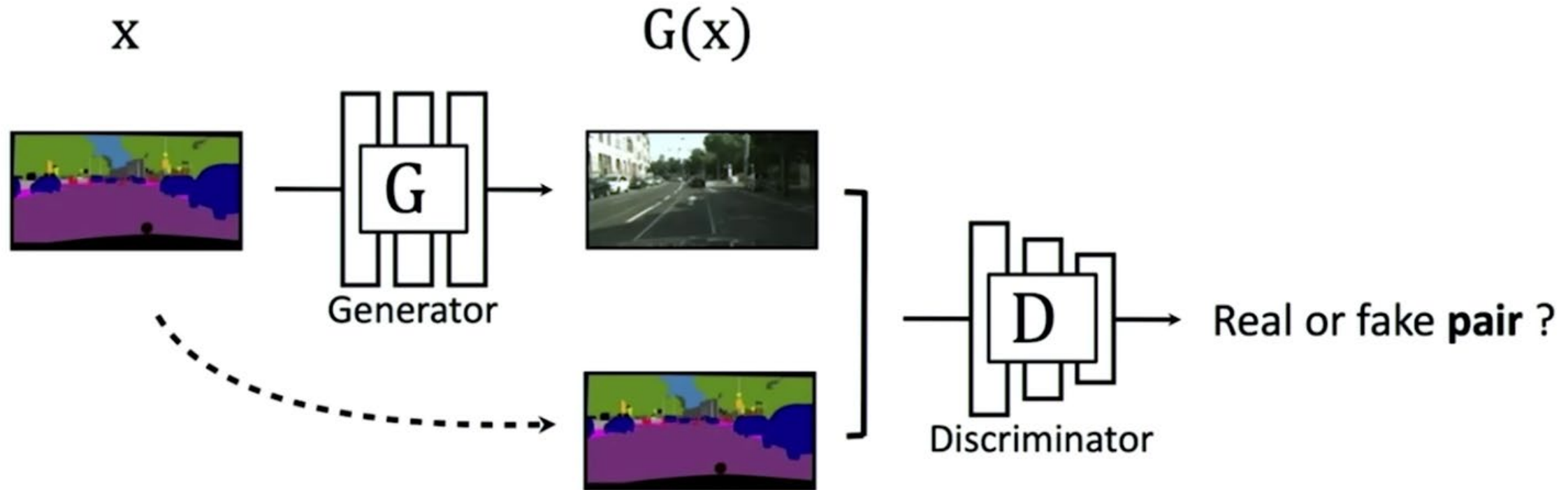


Conditional GANs

What if we want to control the nature of the output, by **conditioning** on a label?



Conditional GANs and pix2pix: paired translation



The discriminator, D , classifies between fake and real **pairs**.
The generator, G , learns to fool the discriminator.

Applications of paired translation

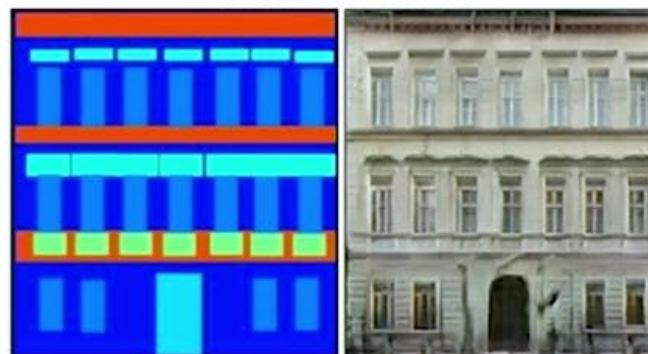
Labels to Street Scene



input

output

Labels to Facade



input

output

BW to Color



input

output

Aerial to Map



input

output

Day to Night



input

output

Edges to Photo

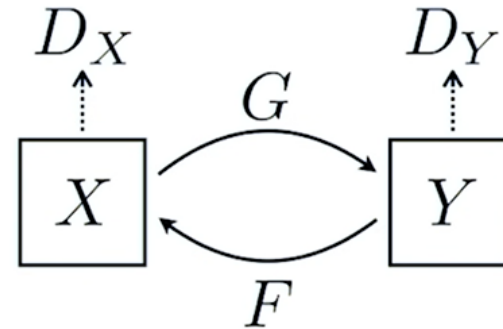


input

output

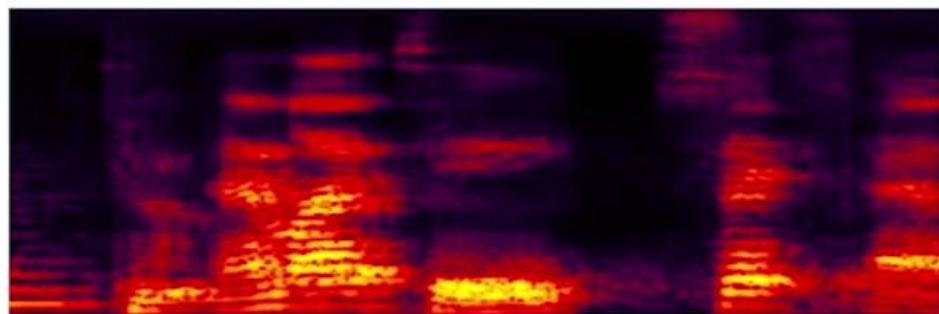
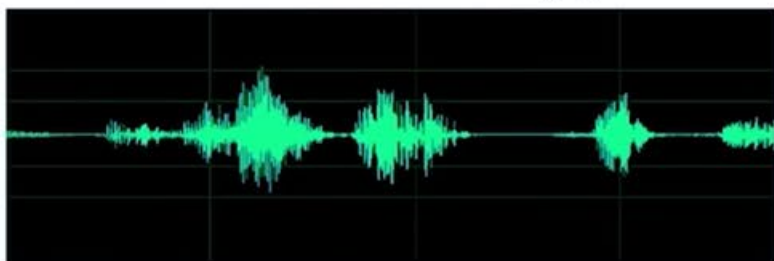
CycleGAN: domain transformation

CycleGAN learns transformations across domains with unpaired data.



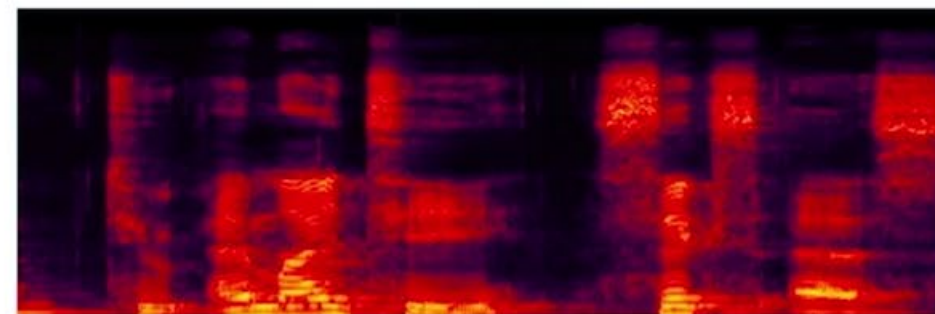
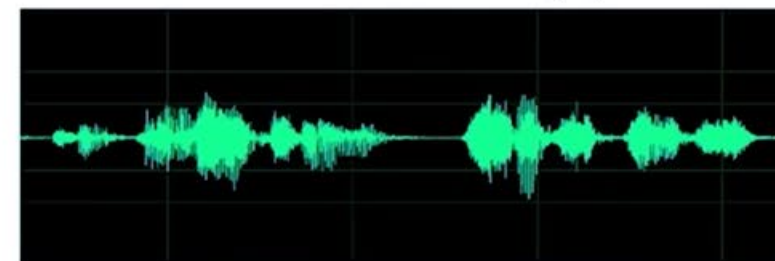
CycleGAN: transforming speech

Audio waveform (A)

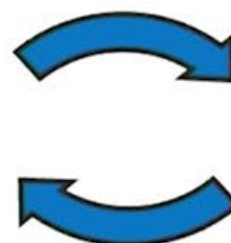


Spectrogram image (A)

Audio waveform (B)



Spectrogram image (B)





Lab5

- Transcript Dataset
 - Augmentation and labeling Dataset project (challenge)
 - เป็น link folder ภาพที่จำกัดสิทธิการเข้าถึงเฉพาะคนในกลุ่มและอาจารย์
- Curriculum books
 - แมพหน้า curriculum books ของแต่ละหลักสูตร กับ ground truth ที่ให้
 - เป็นไฟล์ csv บอกว่า ground truth ชื่ออะไรหลักสูตรไหน เอาข้อมูลมาจากเล่มหลักสูตรหน้าไหนบ้าง
 - คำถาม ที่สามารถหาคำตอบได้จาก ground truth ดังกล่าว พร้อมคำตอบที่ระบุหน้าที่เป็นข้อมูลของหลักสูตรนั้น และข้อของข้อบังคับ (ถ้ามี)

ส่งในกลุ่ม discord ก่อนวันศุกร์ที่ 31 กค. เวลา 6am.